SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO2: Apply suitable Data Preprocessing techniques like Data cleaning, Data intergration, Data transformation and data Reduction, for effective data preparation (BL3, BL4)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B

Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _M. Pranay (192525382) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *M. Pranay*

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B

Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.

Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi)	**Claims per 100 Policyholders (yi)**
3.2	15
4.0	18
5.5	25
4.8	20

---

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):
110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

---

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**.

Sample Data Set (Sorted BMI):
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

---

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**.
To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):
120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

---

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:

(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

ANSWERS

1. Covariance Analysis in Health Insurance

Given data:

Premium increase (xi)	Claims (yi)
3.2	15
4.0	18
5.5	25
4.8	20

Mean of x: $\bar{x} = (3.2 + 4.0 + 5.5 + 4.8) / 4 = 17.5 / 4 = 4.375$.

Mean of y: $\bar{y} = (15 + 18 + 25 + 20) / 4 = 78 / 4 = 19.5$.

Population covariance: $Cov(x,y) = \Sigma [(xi - \bar{x}) (yi - \bar{y})] / n$.

xi	yi	$xi - \bar{x}$	$yi - \bar{y}$	Product
3.2	15	−1.175	−4.5	5.2875
4.0	18	−0.375	−1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125
Total				12.25

Cov(x,y) = 12.25 / 4 = 3.0625.

Interpretation: Covariance is positive, so premium increase and claim frequency have a positive relationship. As premium increase rises, claim frequency tends to increase.

Note: If sample covariance is required, divide by $n - 1 = 3$; the result is 4.0833. The relationship remains positive.

2. Data Smoothing Using Equal-Frequency Binning

Given 27 sorted systolic blood pressure readings. Divide the data into 3 equal-frequency bins of 9 values each and smooth each bin using its mean.

Bin 1: 110, 112, 115, 115, 118, 120, 120, 122, 124. Mean = 1056 / 9 = 117.33. Smoothed values: 117.33 repeated 9 times.

Bin 2: 124, 126, 126, 126, 126, 130, 135, 135, 138. Mean = 1166 / 9 = 129.56. Smoothed values: 129.56 repeated 9 times.

Bin 3: 138, 138, 138, 140, 145, 150, 152, 160, 180. Mean = 1341 / 9 = 149. Smoothed values: 149 repeated 9 times.

Result: Equal-frequency binning reduces minor fluctuations and makes the overall trend easier to analyze.

3. Equal-Width Binning for Risk Categorization (5 Marks)

Given sorted BMI values: 18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5.

Range = 39.5 − 18.5 = 21. Assuming 3 equal-width bins, bin width = 21 / 3 = 7.

Bin	BMI values	Smoothed values (nearest boundary)
1	18.5, 19.2, 21.8, 24.9, 25.5	18.5, 18.5, 18.5, 25.5, 25.5
2	26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4	25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5
3	33.1, 33.6, 34.7, 38.2, 39.5	32.5, 32.5, 32.5, 39.5, 39.5

Result: Equal-width binning divides BMI values into intervals of equal width and replaces values with the nearest bin boundary.

Note: The question does not specify the number of bins; this answer assumes 3 bins.

4. Data Reduction Using Bin Medians (Medical Supply Costs)

Sorted costs: 120, 140, 145, 145, 148, 150, 150, 150, 155, 155, 155, 155, 160, 160, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300.

There are 27 values. Divide into 3 equal-frequency bins of 9 values each.

Bin 1: 120, 140, 145, 145, 148, 150, 150, 150, 155. Median = ₹148. Replace all values in the bin with ₹148.

Bin 2: 155, 155, 155, 155, 160, 160, 165, 170, 170. Median = ₹160. Replace all values in the bin with ₹160.

Bin 3: 175, 175, 175, 175, 180, 190, 200, 210, 230. Median = ₹180. Replace all values in the bin with ₹180.

Result: Bin-median smoothing reduces variation in medical supply costs and simplifies the data for mining.

Note: The supplied data contains 27 values; the last value ₹300 is omitted from the three 9-value bins above. If all values must be included, the bins need to be adjusted.

5. Standardization of Patient Age Data

Given ages: 18, 22, 25, 42, 28, 43, 33, 35, 56, 28.

Mean: $\bar{x} = 330 / 10 = 33$ years.

Absolute deviations from the mean:

Age (xi)	$\lvert xi - 33 \rvert$
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5
Total	88

$MAD = \Sigma|xi - \bar{x}| / n = 88 / 10 = 8.8$ years.

Population standard deviation for z-scores: $\Sigma(xi - \bar{x})^2 = 1174$; $\sigma = \sqrt{(1174 / 10)} = \sqrt{117.4} \approx 10.835$.

Age	Calculation	Z-score
18	(18 − 33) / 10.835	−1.38
22	(22 − 33) / 10.835	−1.02
25	(25 − 33) / 10.835	−0.74
42	(42 − 33) / 10.835	0.83

Final answers: Mean age = 33 years; MAD = 8.8 years; first four z-scores = −1.38, −1.02, −0.74, 0.83.

Interpretation: Negative z-scores indicate ages below the mean; a positive z-score indicates an age above the mean.